

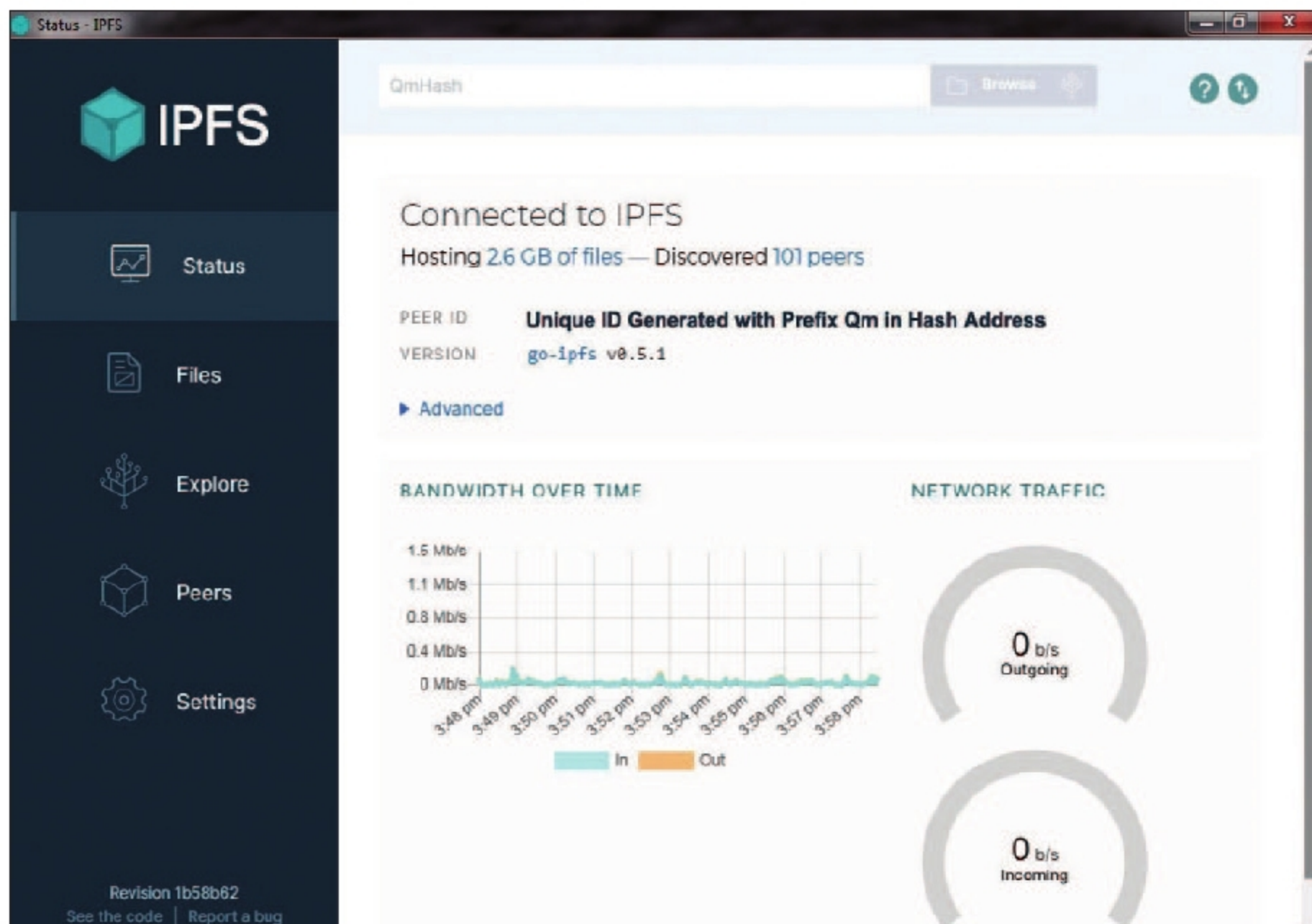


Figure 4: Dashboard and status of IPFS

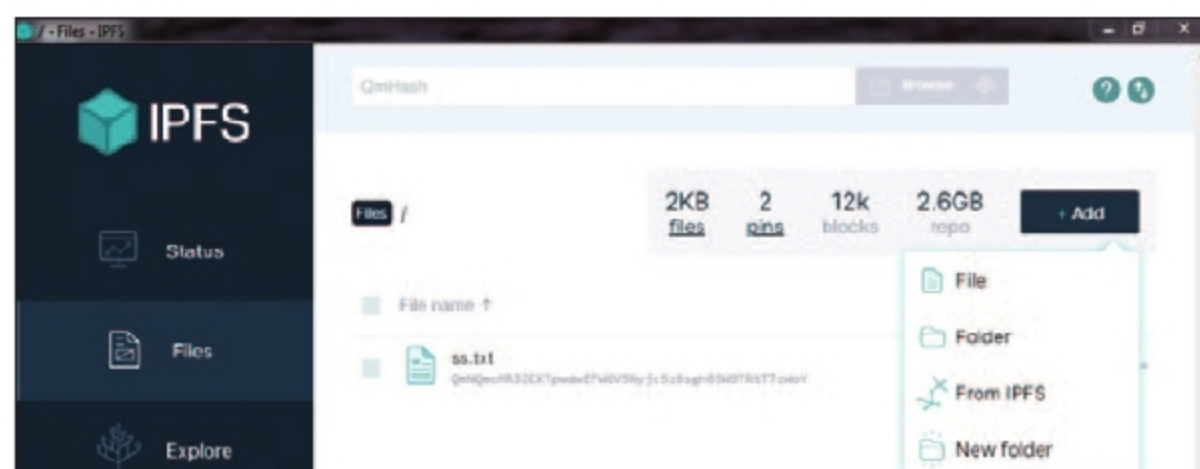


Figure 5: Uploading and integrating files in IPFS

secured and all the addresses are encrypted. As shown in Figure 5, the address of file *ss.txt* has a unique encrypted hash address with prefix *Qm*. The remaining part of the address denotes the assorted properties of the file. Communication in blockchain using IPFS is thus encrypted and peer-to-peer communication is achieved with higher integrity and security.

Scope for research and development

There is huge scope for research in blockchain technologies for scholars, professors and practitioners. As

e-governance services are the most commonly targeted by hackers, crackers and sniffers, these platforms can be secured using advanced decentralised cloud technologies. Performance evaluation and comparison with traditional Web environments can be done by deploying these services on decentralised clouds. In addition, this type of implementation reduces the scope for corruption in Web based services because there is no single administrator and all the communications are transparent, with no possibility for manual or accidental modifications in the records. **END** 🐧

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